

63rd ASH Annual Meeting Abstracts

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616.ACUTE MYELOID LEUKEMIAS: INVESTIGATIONAL THERAPIES, EXCLUDING TRANSPLANTATION AND CELLULAR IMMUNOTHERAPIES

A Phase 3 Randomized Controlled Trial (MICROCARE) to Evaluate the Efficacy of DAV132 in Preventing *Clostridioides Difficile* Infection in Patients with Newly Diagnosed Acute Myeloid Leukemia or High-Risk Myelodysplastic Syndrome and Treated with Intensive Chemotherapy

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Abstract Background:

Acute Myeloid Leukemia (AML) and high-risk Myelodysplastic Syndrome (MDS) patients frequently receive antibiotics (ABX) for febrile neutropenia (prophylaxis or empirical treatment) or for suspected or documented bacterial infections. ABX affect the composition of the human intestinal microbiome and the resulting dysbiosis can have important consequences for patients including *Clostridioides difficile* infections (CDI). The reported cumulative incidence of CDI in patients with AML varies from 9 to 24% in the first 120 days following the start of induction chemotherapy and from 8 to 31% after allogeneic Hematopoietic Stem Cell Transplant (allo-HSCT). In patients undergoing allo-HSCT, acute Graft-versus-Host Disease (GvHD) and GvHD-related mortality are other important complications associated with intestinal dysbiosis, as shown in experimental studies in mice and retrospective analyses in humans.

DAV132 is a microbiome protective therapy that can be administered concomitantly with oral or intravenous ABX. It has been primarily developed for adsorbing antibiotic residues in the distal ileum and colon in order to preserve the integrity of the gut microbiome, without impacting antimicrobial pharmacokinetics. Randomized clinical trials conducted to date, both in healthy volunteers and in patients, showed that DAV132 adsorbed residues of ABX in the colon and protected the intestinal microbiome from the dysbiosis induced by ABX while preserving their bioavailability. In addition, DAV132 was safely used in hospitalized patients receiving (multiple) concomitant drugs.

Study Design and Methods:

MICROCARE is a multicenter, randomized, placebo-controlled, parallel-arm phase 3 clinical trial. A total of 900 patients (randomized 1:1 to either DAV132 or placebo) are planned to be enrolled in approximately 80 study sites worldwide.

The primary objective of this study is to evaluate the efficacy of DAV132 in preventing CDI in patients with newly diagnosed AML or high-risk MDS treated with intensive chemotherapy. In addition, the study aims to determine whether the protection of the intestinal microbiome with DAV132 can reduce the incidence of colonization with multidrug-resistant bacteria, bacteremia, and acute GvHD, and improve the long-term outcomes of transplanted patients.

Randomization will be stratified by (i) site, and (ii) use of ABX within 30 days before inclusion and/or history of CDI. During the cycle of induction chemotherapy, DAV132 will be administered to patients every day until neutrophil recovery, regardless of the administration of ABX treatment. After neutrophil recovery and up to day 120, DAV132 will be administered every time an ABX treatment is given, for the length of the ABX treatment plus 2 days.

The primary analysis of this study is the comparison of the time to CDI between DAV132 patients versus placebo patients up to 120 days after randomization. This will be analyzed using the cause-specific Cox proportional hazards model, whereby time to CDI is the outcome of interest, and mortality will be considered as competing event. In the primary outcome, a cause-specific Cox model will be used assuming that the impact of DAV132 on mortality is negligible; this will be confirmed in a secondary analysis. Secondary and exploratory endpoints include the rate of patients with ABX-associated diarrhea, the rate of patients with a gut colonization by resistant bacteria or a bloodstream infection, the diversity of the gut microbiome, the incidence of acute GvHD and overall survival. Two interim analyses will be performed at approximately 35 and 50 CDI events.

Study status: Recruitment of patients in the study is starting in multiple countries in Europe. A first patient has been randomized in July 2021.

Topics: Microbiome; CDI; GvHD; Dysbiosis; Antibiotics

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